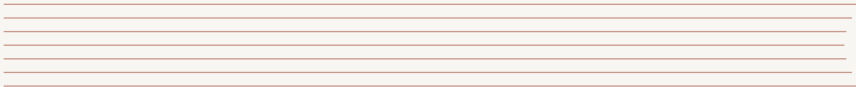


POWERING THE NATION: CAN YOU CONTROL A NUCLEAR REACTOR?

An Introduction to Reactor Safety & Dynamics

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THE BIG QUESTION: POWERING SINGAPORE'S FUTURE

Singapore faces a major challenge known as the **Energy Trilemma**:

Secure Can we guarantee
a stable supply that never
fails?

Affordable Can we keep
electricity costs low for
everyone?

Sustainable Can we
generate power without
contributing to climate
change?

■ A Potential Solution?

Advanced nuclear reactors are being considered globally as a high-tech solution that could potentially address all three points. But are they safe?

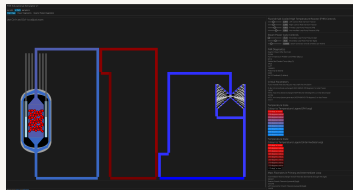
YOUR MISSION, SHOULD YOU CHOOSE TO ACCEPT IT

Today, you are not just students. You are reactor operators at the controls of a next-generation Fluoride-salt-cooled High-temperature Reactor (FHR).

The Situation: The grid operator requests an immediate increase in power output to meet a sudden demand spike.

■ Your Objective

Increase reactor thermal power from **30 MWth** to **60 MWth** and stabilize it there.



OPERATIONAL CONSTRAINTS & SAFETY LIMITS

Nuclear reactors are powerful systems that require precise control. We cannot just "floor the accelerator."

The Rules of Engagement:

- ⦿ You have full control over the reactor's control rods (reactivity).
- ⦿ You must aim for exactly **60 MWth**.

■ SAFETY LIMIT (FAILURE CONDITION)

If reactor power exceeds **1000 MWth** at any point during the maneuver, you have violated the reactor's safety limits and failed the mission.

YOUR INTERFACE: THE FHR CONTROL PANEL

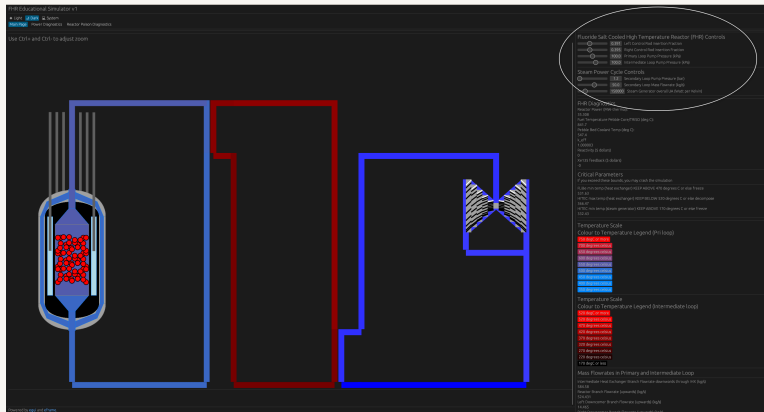


Figure: Current State: Stable at 30 MWth

THE CHALLENGE: SOLVE IT AT YOUR DESK

You've seen the controls and you know the mission. Now it's your turn to try and crack the problem.

■ Your Task (5 Minutes)

On your own computer, launch the simulator and attempt the power increase from **30 MWth to 60 MWth**.

- ⦿ Your goal is to find a strategy to hit the target and stabilize the power.
- ⦿ Pay close attention to how much you move the controls and for how long.

THE CHALLENGE: SOLVE IT AT YOUR DESK

■ Think You've Cracked It?

If you figure out a successful technique, get ready! In a few minutes, I will be asking for volunteers to come up and demonstrate their solution on the main screen in front of the class.

Your time starts now!

Volunteers Needed!

Who has the steady hand to bring us to 60 MWth without exceeding 1000 MWth?

For the rest of the class (Observers): Watch closely. Don't just look at the power level.

- ◉ How does the reactor respond immediately after they move the controls?
- ◉ What happens to the power when they *stop* moving the controls?
- ◉ Does the reactor feel "heavy" (slow to move) or "twitchy" (fast to move)?

DEBRIEF: INTUITION VS. REALITY

It wasn't as easy as turning up a volume knob on a radio.

Discussion Questions:

- ① Did anyone overshoot the 60 MWth target? Why do you think that happened?
- ② Did you notice the power continuing to change even after the controls were put back to neutral?
- ③ If you added too much reactivity, did it feel like the reactor was "running away" from you?

Why does the reactor have "momentum"? Why doesn't it stop instantly?

To answer this, we need to understand what makes nuclear engineering a unique challenge.

IS IT REALLY THAT SIMPLE?

On paper, the concept of a nuclear reactor sounds straightforward:

"Neutrons split uranium to heat something that then boils water to spin a turbine."

But you just experienced it for yourselves. It wasn't that simple.

- ◉ The power had "momentum."
- ◉ It was easy to overshoot the target.
- ◉ It didn't respond like a car engine or a volume knob.

IS IT REALLY THAT SIMPLE?

I The Core Question

If the basic idea is so easy, why is controlling it so tricky?

The gap between that simple sentence and a safe, working power plant is precisely why nuclear science and engineering is such a deep and fascinating field.

THE JOURNEY FROM PAPER TO POWER PLANT

To use nuclear energy to help solve Singapore's energy trilemma, we must bridge that gap.

How do we get from a paper design to a safe, working plant that can operate for 60+ years?

It's a grand challenge with many layers:

- ◉ **Politics & Public Policy:** How do we regulate and govern this technology?
- ◉ **Economics & Finance:** How do we fund and build such a large project?
- ◉ **Social Acceptance:** How do we ensure public trust and communicate effectively?
- ◉ **Technical & Engineering Safety:** How do we guarantee the plant will be safe under all possible conditions?

I Our Focus Today: Technical Safety

My work, and what we'll explore next, lives in that last box. This is the world of **Reactor Safety**, where we must understand every physical process inside the core.

WHAT PHENOMENA DO WE STUDY?

To guarantee safety, we can't just have a good design on paper. We have to understand, predict, and master every single physical process happening inside the reactor.

We call these the core **phenomena**.

■ A Tour of the Frontier

Let's take a quick tour of the four most critical areas that reactor safety engineers and scientists spend their careers studying.

PHENOMENON #1: NEUTRONICS - THE ENGINE OF THE REACTOR

Concept: Neutronics is the study of the neutron population in the reactor. This population directly determines the reactor's power level, generating about 90% of the heat during operation.

The Safety Question: How do we precisely predict and control the neutron population to generate steady power and ensure the chain reaction never runs away?

PHENOMENON #1: NEUTRONICS - THE ENGINE OF THE REACTOR

I From Simple Models to Reality

For basic control dynamics—the “momentum” you felt—we can use the **Point Reactor Kinetics Equations (PRKE)**.

- ⦿ It simplifies the entire reactor down to a single point in space.
- ⦿ **The FHR simulator you used runs on the 6-group PRKE model** to accurately capture the effect of delayed neutrons.

However, true neutron behavior is far more complex. To design a reactor, we must account for:

- ⦿ **Where** neutrons are in the reactor (Spatial effects)
- ⦿ **What direction** they are traveling (Angular effects)
- ⦿ **How fast** they are moving (Energy effects)

This is all described by the Neutron Transport Equation, a much more complicated model that forms the basis of modern reactor physics.

PHENOMENON #2: DECAY HEAT - THE FIRE THAT WON'T GO OUT

Concept: The other 10% of the reactor's heat comes from a different source. Even after you completely stop the chain reaction, the core continues to produce heat from the radioactive decay of its waste products.

Think of it like a cooker stove with a unique problem: you can turn the knob to "off" to stop the main flame, but you cannot turn off the burner completely. It continues to generate heat no matter what.

■ A Real-World Example: Fukushima

The reactors at Fukushima Daiichi shut down perfectly when the earthquake hit. The meltdowns were caused by the failure to remove this powerful **decay heat** after the tsunami destroyed the cooling pumps.

PHENOMENON #2: DECAY HEAT - THE FIRE THAT WON'T GO OUT

The Safety Question: How do we design systems that can remove this residual heat reliably, even if all power is lost? This leads us directly to our next challenge...

PHENOMENON #3: HEAT TRANSFER - THE COOLING CHALLENGE

Concept: Heat transfer is about moving heat from where it's produced (the fuel) to where it can be used (the turbine). This involves two major challenges.

■ Problem 1: Avoiding Overheating (in Water-Cooled Reactors)

In Pressurized and Boiling Water Reactors (PWRs/BWRs), if the fuel gets too hot, the water can form a steam blanket around it (Critical Heat Flux). This blocks cooling and can lead to fuel damage.

PHENOMENON #3: HEAT TRANSFER - THE COOLING CHALLENGE

■ Problem 2: Ensuring Coolant Flow (in All Reactors)

To remove decay heat and prevent a Fukushima-like event, we must keep the coolant circulating.

- ⦿ **Forced Circulation:** Uses pumps. Powerful and controllable, but requires electricity.
- ⦿ **Natural Circulation:** Uses physics (hot coolant rises, cool coolant sinks). It is passive and highly reliable, operating without any pumps or external power.

Modern advanced reactors are designed to have strong natural circulation capabilities as a core safety feature.

Note: For HTGRs, FHRs, and MSR, which use single-phase coolants (gas or liquid salt), Critical Heat Flux is not a concern, allowing them to operate at higher temperatures and focus on the circulation problem.

CASE STUDY: TESTING HEAT TRANSFER WITHOUT A REACTOR

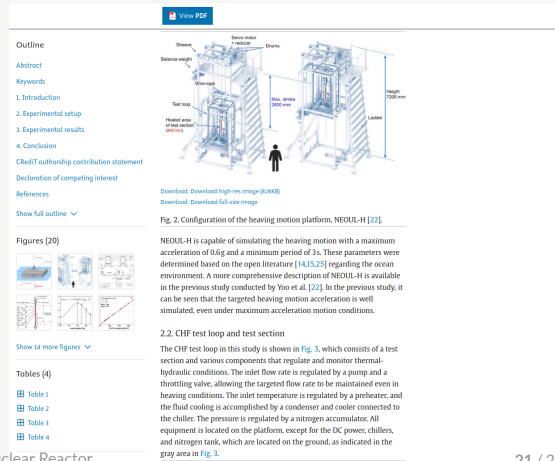
To study complex phenomena like Critical Heat Flux, we don't need a real reactor. We can build sophisticated non-nuclear test facilities.

How It Works

Instead of nuclear fuel, we use precisely machined **electric heater rods**.

- These rods can be powered to perfectly mimic the heat output of actual fuel.
- This is far safer, cheaper, and allows for detailed measurements.
- We can study complex effects, like how the rocking motion of a ship might affect CHF on a marine reactor

Ku et al. 2026.



CASE STUDY: TESTING NATURAL CIRCULATION WITH CIET

For system-wide behavior like natural circulation, we build "Integral Effects Test" (IET) facilities that model the entire reactor loop.

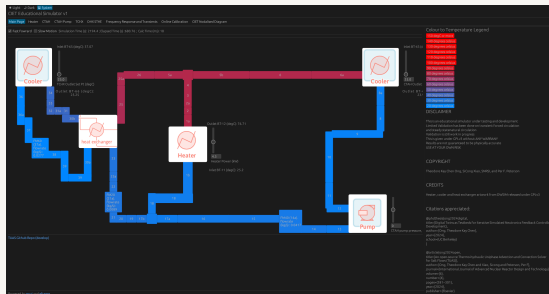


Figure: The Compact Integral Effects Test (CIET) Educational Simulator.

The Limits of Natural Circulation

It is a powerful passive safety feature, but it has challenges:

- **Lower Flow Rate:** The flow is inherently weaker than with pumps, so heat removal is slower.
- **Flow Instability:** The flow can become unstable and oscillate, which can hamper heat removal.
- **CHF is Still a Risk:** In water systems, we must prove that the weaker natural flow is still fast enough to

PHENOMENON #4: MATERIALS SCIENCE - FORGING THE UNBREAKABLE

Concept: The inside of a reactor is one of the most extreme environments humans have ever created: intense radiation that damages atomic structures, high temperatures, and high pressure.

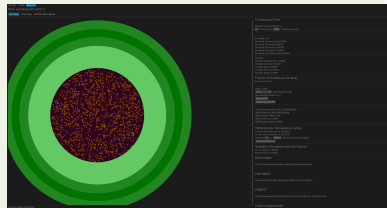
The Safety Question: What can we possibly build the reactor out of? How do we invent and test materials that can perform their job flawlessly for over 60 years without cracking, corroding, or becoming brittle?

PHENOMENON #4: MATERIALS SCIENCE - FORGING THE UNBREAKABLE

■ An Example of Extreme Engineering: TRISO Fuel

Advanced reactors like the **High-Temperature Gas-Cooled Reactor (HTGR)** and the **Fluoride-salt-cooled High-temperature Reactor (FHR)** use TRISO fuel.

Each particle, the size of a poppy seed, contains a uranium kernel wrapped in multiple layers of incredibly durable ceramic. These layers act as a tiny, personal containment vessel, trapping radioactive products inside even at extreme temperatures up to 1600°C.



Cross-section of a TRISO particle.